

MY SPICE

INDUSTRIAL DESIGN
SENIOR PROJECT 2024

CREATED CARLOS OCASIO MILANES

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INITIAL METHODOLOGY PLAN

1. Find Problem
2. Research about the problems and branch out
3. Conduct interviews and ask questions
4. Research current market solutions
5. Generate Ideas and start sketching
6. Collect Data run tests and surveys and analyze
7. Make Prototypes
8. Synthesize Ideas
9. Make Models
10. Evaluate

PROBLEM

Kitchen Area Struggles

01. RESEARCH

QUOTE

“my wife left me because I left dishes at the sink”

TOP PROBLEMS

- Not enough counterspace
- Greasy Surface
- Clutter
- Not Having enough time
- Insufficient storage





RESEARCH

Not enough counter space

Not having enough space in the kitchen is a worldwide problem that comes from there being many kitchen appliances that are useful. A big part of it also comes from there being clutter in the kitchen. Popular solutions to this problem include using your cabinets and hanging utensils on the wall.

Greasy Surfaces

When cooking you use a lot of ingredients, utensils and cooking appliances that can all get pretty fiery and messy quickly. This mess easily spreads around the kitchen and can rapidly become a problem, spreading grease, mess and food odors around the whole kitchen. Some of the solutions for this problem include spraying cleaning solutions and using kitchen cloths to deep clean your kitchen, which in turn gives you another problem to solve.

Clutter

A pervasive issue in kitchens worldwide, stemming from the abundance of utensils, ingredients, and appliances essential for cooking. The resulting disarray can impede efficiency and make meal preparation a daunting task. To tackle clutter, many resort to utilizing cabinets for storage and employing wall-mounted racks for hanging utensils, optimizing available space and creating a more organized cooking environment.

RESEARCH

Not Having Enough Time

This is a very common challenge in the kitchen, particularly for busy individuals juggling various responsibilities. With limited time to devote to meal preparation, cooking can feel rushed and stressful. To address this issue, some opt for meal prepping strategies, such as batch cooking or planning quick and easy recipes, streamlining the cooking process and ensuring meals are ready in a timely manner.

Insufficient Storage

This is a problem that plagues many kitchens, leaving little room to neatly store cookware, pantry items, and kitchen gadgets. This lack of space can lead to clutter and make it challenging to keep the kitchen organized and functional. To mitigate the problem of insufficient storage, homeowners often invest in storage solutions like shelves, racks, and pantry organizers, maximizing available space and creating a more efficient cooking environment.



CURRENT SOLUTIONS

Meal Delivery Services

Meal delivery services like DoorDash offer convenience, but they often come with downsides. While they provide quick access to a variety of food options, the cost can add up, especially with delivery fees and tips. Moreover, these services primarily deliver from fast-food chains and restaurants, which may not always offer the healthiest options, leading to potential overconsumption of unhealthy foods.

Food Prep Delivery

HelloFresh and similar meal plan services provide pre-packaged ingredients and recipes for home cooking. However, they have their drawbacks. Despite the convenience of having ingredients delivered to your doorstep, these services can be pricey, especially when compared to grocery shopping. Additionally, some users find that the meal options become repetitive over time, limiting culinary variety. Furthermore, the quality of ingredients may not always meet expectations, raising concerns about freshness and taste.

Eating Out

While dining out at restaurants offers a break from cooking and cleaning, it's not without its drawbacks. Eating out frequently can strain budgets, as restaurant meals tend to be more expensive than home-cooked ones. Additionally, the time spent traveling to and waiting for meals in restaurants can be significant, especially during peak hours. Moreover, restaurant dishes often contain hidden fats, sugars, and sodium, contributing to unhealthy eating habits over time.

EXPENSIVE SOLUTIONS

Statistics around the Midwest

States	(Out - At Home)	Eating Out	Eating at Home
Illinois	$16.15	$20.00	$3.85
Indiana	$10.50	$15.00	$4.50
Iowa	$11.23	$15.50	$4.27
Kansas	$8.42	$12.00	$3.58



Money you would save by eating at home

https://www.topnutritioncoaching.com/blog/cost-of-eating-out-vs-eating-in

THE EFFECTS

The way the current population is approaching food is very scary. Eating unhealthy food which largely comes from eating out at restaurants that focus on profits more than health and their actual food has caused very serious repercussions.

39.6% Of the US Population are obese with ≈ 300,000 deaths per year, making it the second leading cause of preventable death.

A large part of this comes from the overconsumption of over processed unhealthy foods.

Eating out means you don't have control of the ingredients, portions or nutritions that are put in your food, while eating at home allows you to take control of the things that you put into your body.





MORE EFFECTS

As if this wasn't enough, this type of unhealthy consumption has more negative side effects in the short run and the long run

Short:

- increased stress levels
- fatigue and decreased energy levels
- difficulty sleeping
- concentration difficulties
- feeling down
- tooth decay

Long:

- type 2 diabetes
- heart-related problems
- overweight and obesity
- osteoporosis
- certain cancers
- depression
- eating disorders

Interview & Findings



Key Interview Questions

> What are some of the inconveniences that you go through when preparing a meal with a recipe?

> Can you walk through step by step of a meal preparation and describe your typical experience when following a recipe on your phone or tablet (or other) while cooking?

> What challenges do you face when handling your (music/recipe/etc) device in the kitchen?

> On a scale of from 1 through 10 how much does meal preparation contribute to being a stress factor in your life?

Problems Identified by Interviewees

> The cooking process can be very cumbersome especially when following a recipe

> Dishes are a very annoying activity that sometimes even makes the interviewee refrain from cooking

> There is no convenient space to place your phone while following a recipe and often the interviewee lost track of where they

My Findings

I found that a big part of the problem is the cleanliness since it is the kitchen area, where we make the stuff we eat, and since there's so much food and ingredients that need to be spread through so many different tools and appliances in a precise way, the cooking process can easily become a difficult task.

This confirmed my hypothesis that the kitchen area needs to be more synergetic one, where things can flow easily, and the more things that you have to touch and spread at the same time the messier the whole experience becomes, physically and emotionally.



02. OPPORTUNITY

WHO?

I recognized that I had to design...

For people that are...



looking for cheap fast food options and can benefit greatly from home cooking



parents who are stressed from the cooking process and wish it was easier



looking to streamline the cooking process and make it a more efficient one

OPPORTUNITY

What can I do?

I started imagining what a more synchronous and flowing kitchen environment would look like with the help of AI, from things like digital cutting boards to projectors setup throughout the kitchen to provide easy recipe viewing the possibilities were endless.



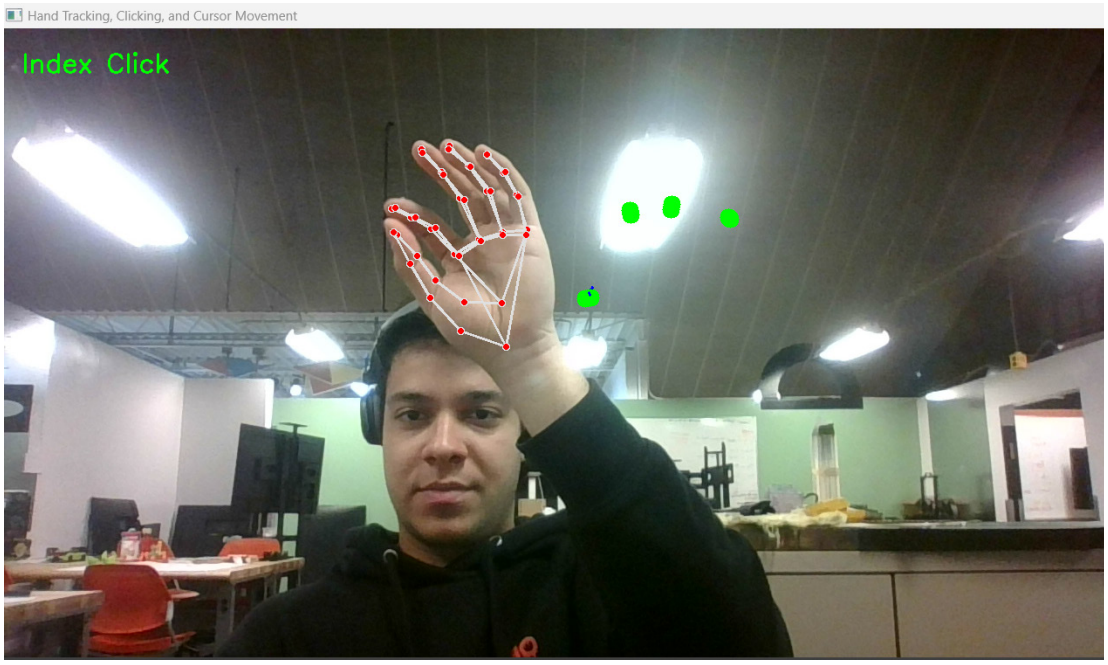


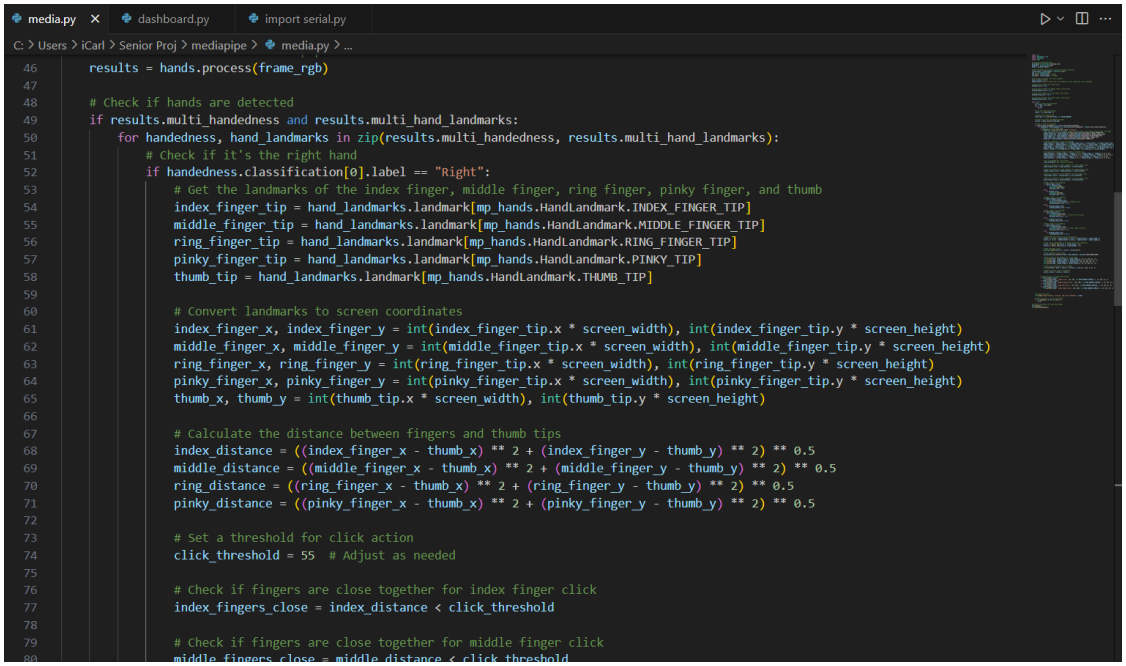


https://gemini.google.com/app/

Hands-Free Technology

At the same time I was working on a hands free, gestured control technology program since from my research I recognized that the mess in the kitchen was one of the biggest problems, and the less things you would have to touch with your dirty hands the less mess. I also designed this technology with recipe viewing and navigation in mind since it is something that we have to do while cooking.





I did this throughout months of work with the use of Visual Studio Code to input the coding and the programming language of python. Since I didn't have any programming experience I also used a lot of help from the Chatbot program ChatGPT which helped me get my ideas and words and convert them into working code.

WHAT NOW?

I knew I wanted to incorporate my hands free algorithm into a synergetic kitchen environment, I just needed a product or way to incorporate it. This led me to looking at the whole kitchen process to see where I could make improvements.



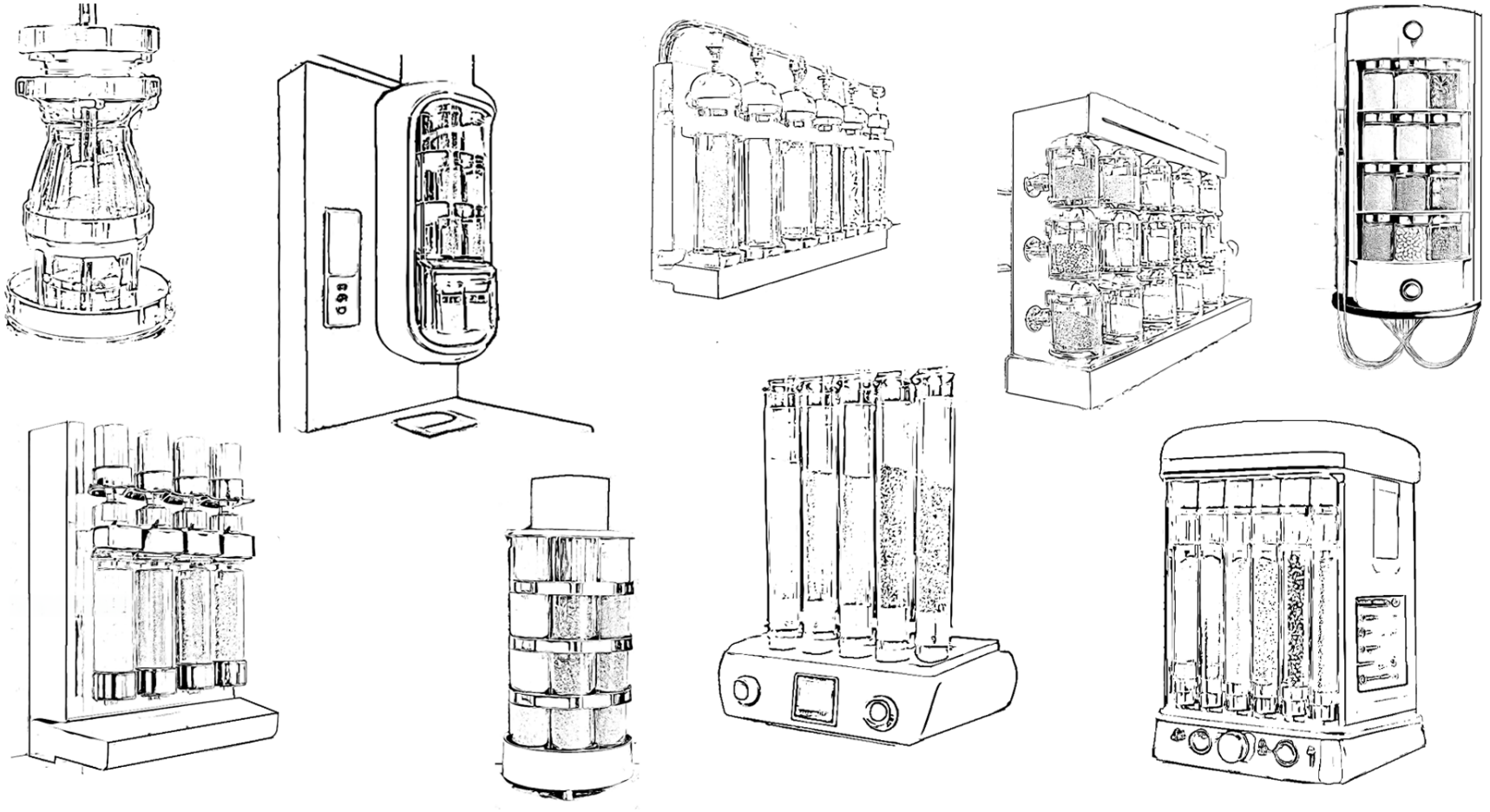
The part I wanted to focus on was when you start cooking, it is the part of the process where you have to keep track of many things at the same time and make sure they go smoothly. I understood that this is the part where we could use the most help

- Plan your meal
  - Thinking about what to make with the time you have, the different appliances you will use and the ingredients that you have available to you.
- Prep your ingredients
  - Taking out the ingredients you will use, chopping them up if needed, defrosting them, and making them readily available to you for when it is cooking time.
- Follow a recipe
  - Choosing a recipe that is to your liking and that it is coalescent with the ingredients and appliances that you have in your kitchen
- Start cooking
  - Here is where the process can vary in many directions depending on your cooking method, but there are a couple of things that apply to all of them, which include seasoning your food, tossing ingredients into cooking appliance, keeping an eye of them and stirring, flipping or moving when necessary.
- Serve and enjoy
  - Plating your food, applying garnish if wanted, getting out your clutery and serving a refreshing drink to sit down and enjoy your food.
- Clean up
  - Putting away your ingredients in the proper manner, cleaning out your dishes and appliances and tossing them into the dishwasher, so that later you can put them away, wiping down your counters and table.

NARROWING DOWN

SPICE DISPENSER

After much consideration and ideation I narrowed down my options to a spice dispenser, spices are something we use usually while we have to do other tasks at the same time, while our hands our dirty. Being able to use spices hands-free and automatively would lead to a more streamlined and clean cooking process. A spice dispenser could also be wall mounted and save counter space and clutter in the kitchen. After all of this I realize it was clear that this was what I should make so I started ideating and sketching with the help of AI.



03. DEVELOPMENT

After I had my idea for a product fleshed out I had to think about the different features and the way I wanted my Spice dispenser to be integrated to the kitchen.

Also a way to integrate some of the current features I already had planned.



Develop an app to utilize and trigger the spice dispenser, as well as add other features that would help throughout the kitchen process.

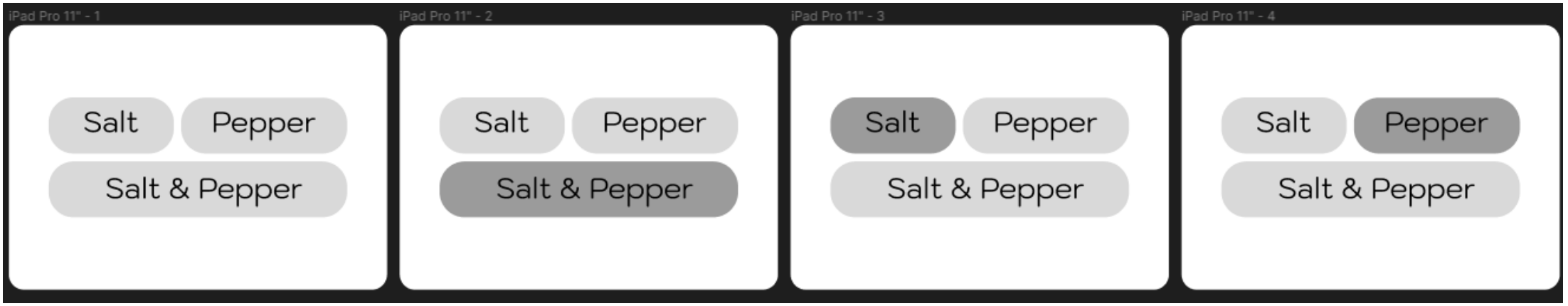


Have a device holder mounted on to the spice dispenser in order to display the recipe and other things the user might need to view.



Incorporate my touch free technology to the device that will be utilized in order to navigate the app without spreading any messes.

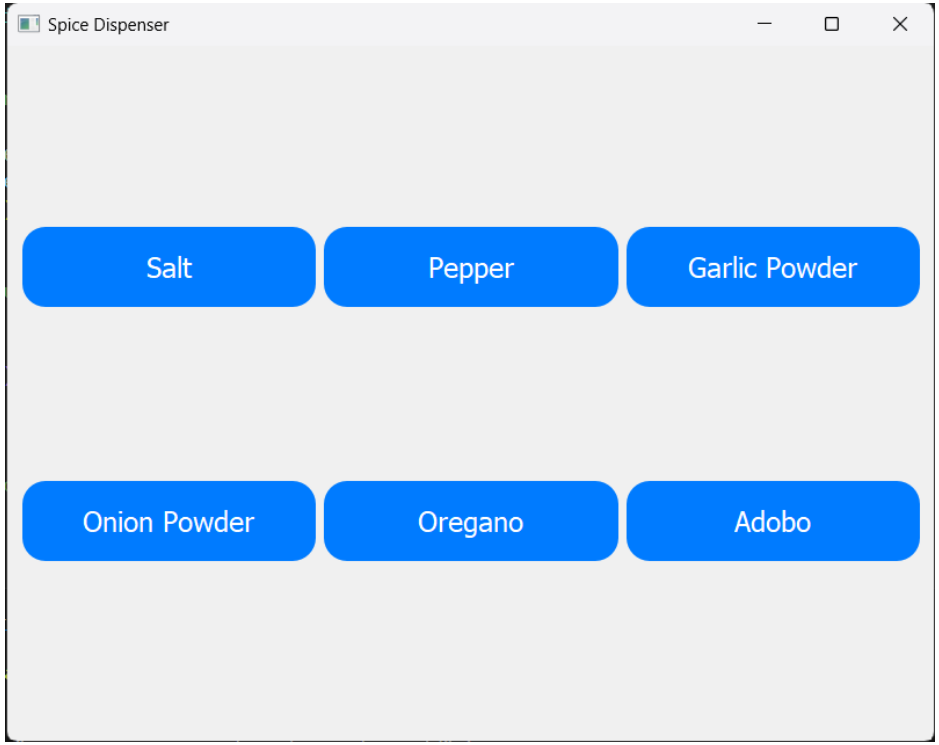
APP



STARTING PHASE

My app development started as a simple way to test how I could connect the digital UX menu to my phyisical product, this was a hard challenge since I hadn't worked with these technologies in the past and I had to figure it out for my product to work.

Little did I know this would be harder than expected and I had to start with a more basic version of UX approach that had to be developed through python in order to be able to control an arduino.



PHYSICAL MODEL

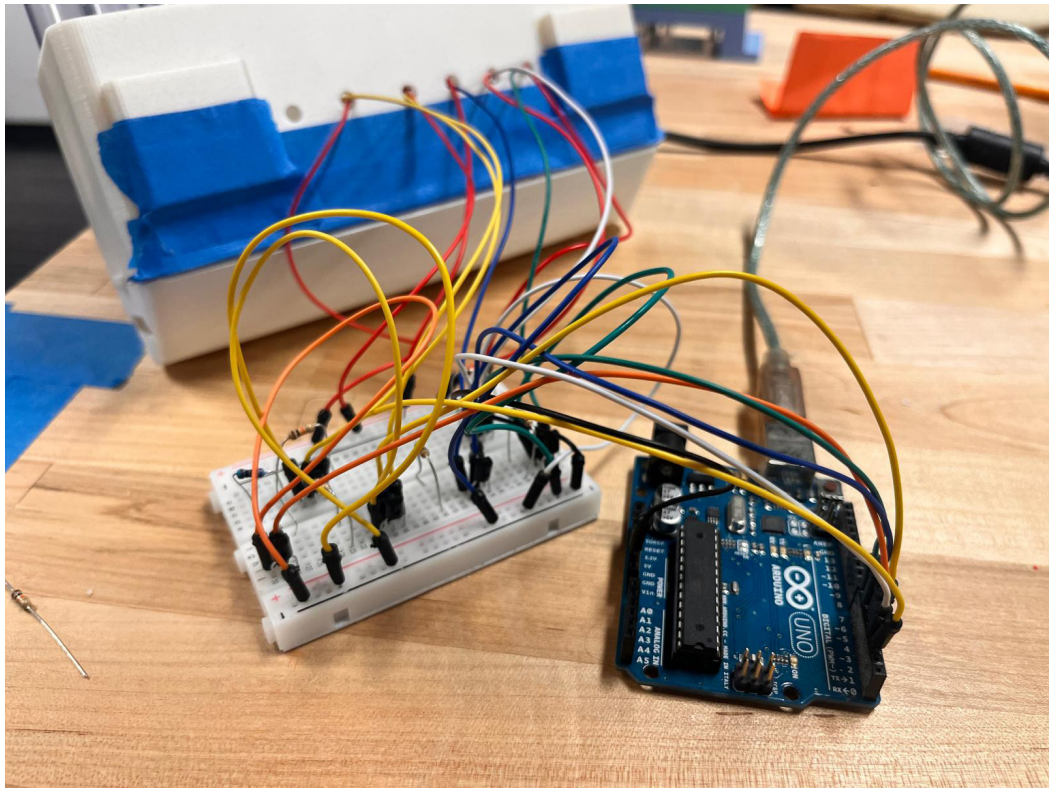
STARTING PHASE

While this was going on I was also working on my physical model. The model consisted of the main base, pipes to hold the spices, device holder and caps and gears that worked together with motors and the arduino to rotate in order for the spice to be able to be dispensed. Through this process I went through a lot of trial and error trying to get the measurements and technology to work correctly.

From learning about transistors, resistors, dc motors, arduino simulation programs, different dispense mechanisms and more I gained a lot of knowledge throughout this process.





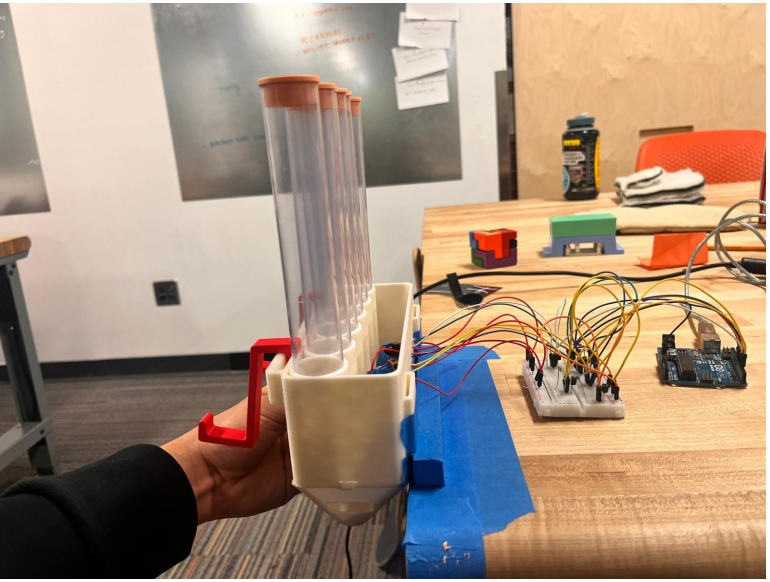


TECHNOLOGY

ARDUINO

Throughout more of the process I inmersed myself in the world of arduinos and technological circuits. After a lot of wiring and tinkering around I designed a code that worked the following way:

You choose which recipe you would like to cook on the UI/UX > This program sends a signal to the python code > This program chooses which motors should have to spin in order to dispense the correct ingredients and amounts > Which communicates with the arduino program > Which communicates with the physical motors > Which turns gears that allow the holes in both caps to line up and dispense the spices



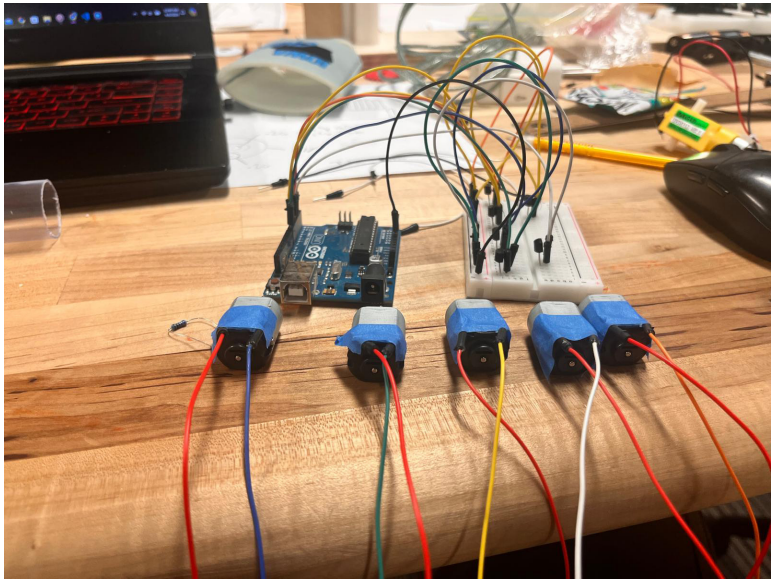


```
sketch_mar25a §
// Arduino code
#define MOTOR_PIN_SALT 2
#define MOTOR_PIN_PEPPER 3
#define MOTOR_PIN_GARLIC 4
#define MOTOR_PIN_ONION 5
#define MOTOR_PIN_OREGANO 6
#define MOTOR_PIN_ADOBO 7

void setup() {
  Serial.begin(9600);
  pinMode(MOTOR_PIN_SALT, OUTPUT);
  pinMode(MOTOR_PIN_PEPPER, OUTPUT);
  pinMode(MOTOR_PIN_GARLIC, OUTPUT);
  pinMode(MOTOR_PIN_ONION, OUTPUT);
  pinMode(MOTOR_PIN_OREGANO, OUTPUT);
  pinMode(MOTOR_PIN_ADOBO, OUTPUT);
}

void loop() {
  if (Serial.available()) {
    String spice = Serial.readStringUntil('\n');
    if (spice == "salt") {
      rotateMotor(MOTOR_PIN_SALT);
    } else if (spice == "pepper") {
      rotateMotor(MOTOR_PIN_PEPPER);
    } else if (spice == "garlic powder") {
      rotateMotor(MOTOR_PIN_GARLIC);
    } else if (spice == "onion powder") {
      rotateMotor(MOTOR_PIN_ONION);
    } else if (spice == "oregano") {
      rotateMotor(MOTOR_PIN_OREGANO);
    } else if (spice == "adobo") {
      rotateMotor(MOTOR_PIN_ADOBO);
    }
  }
```

```
Done uploading.
Sketch uses 1902 bytes (5%) of program storage space. Maximum is 32256 bytes.
Global variables use 184 bytes (8%) of dynamic memory, leaving 1864 bytes for local variables. Maximum is 2048 bytes.
43
```





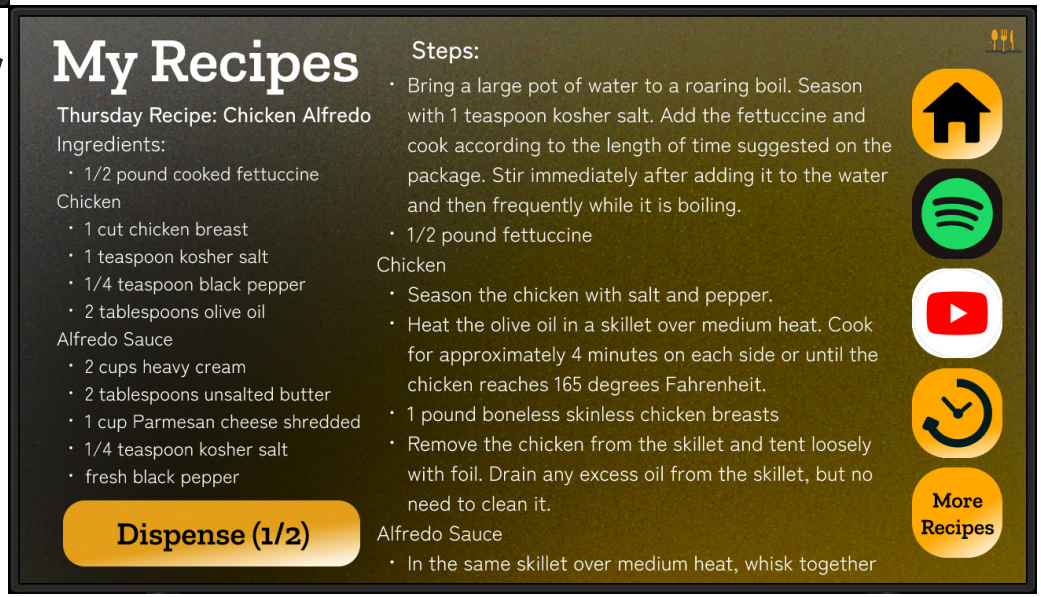
I decided on a basic homescreen with big buttons that could be easily pressed and access with the gesture control features.

I added a big button for the main feature which is the daily planned recipe. Other buttons were added as well, embedded spotify shortcut for music while cooking, youtube as well in case of following video recipes or entertainment. Lastly a timer app that allows you to keep track of the cooking time of the different things in your kitchen.

In the recipes app there are various things to note, there is an ingredients list thats easily accessible and easy to read as well as a scrollable step by step guide on how to follow the recipe.

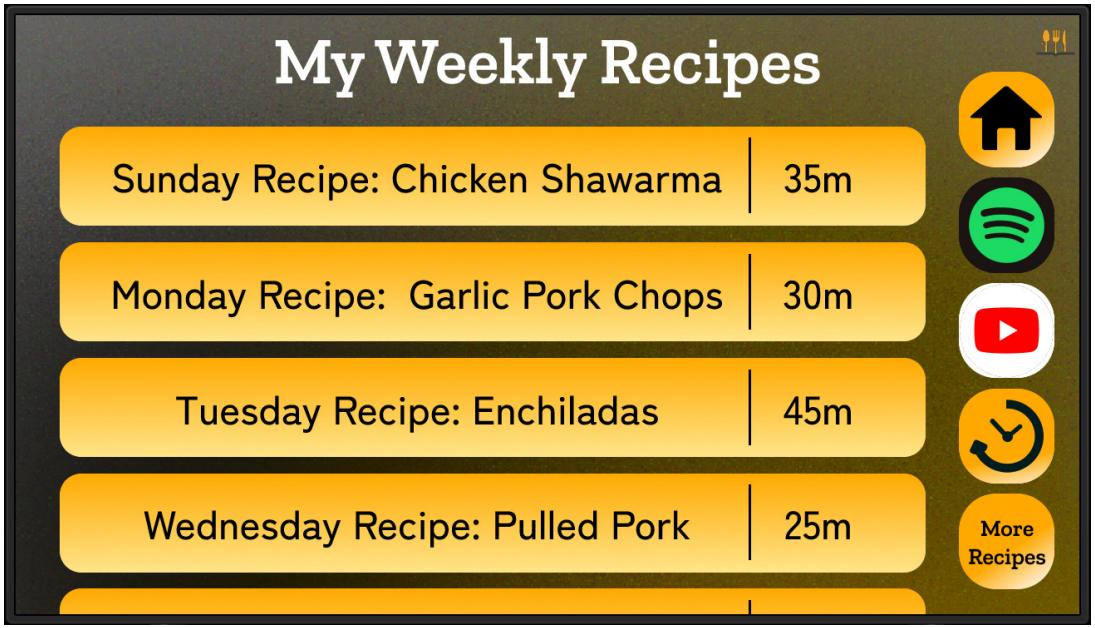
Another important feature here is the dispense buttons which triggers the mechanism so that all of the needed spices for that recipe are all ready for you in the physical spice collector.

Lastly there are all of the homescreen options on the right plus a more recipes button.



04. APP DEVELOPMENT

After figuring out the logistics of the app it was time to develop more of my apps features and what I wanted it to look like.



In the more recipes section you are allowed to see and move around your recipes for the week as well as see the time that it takes to cook them.



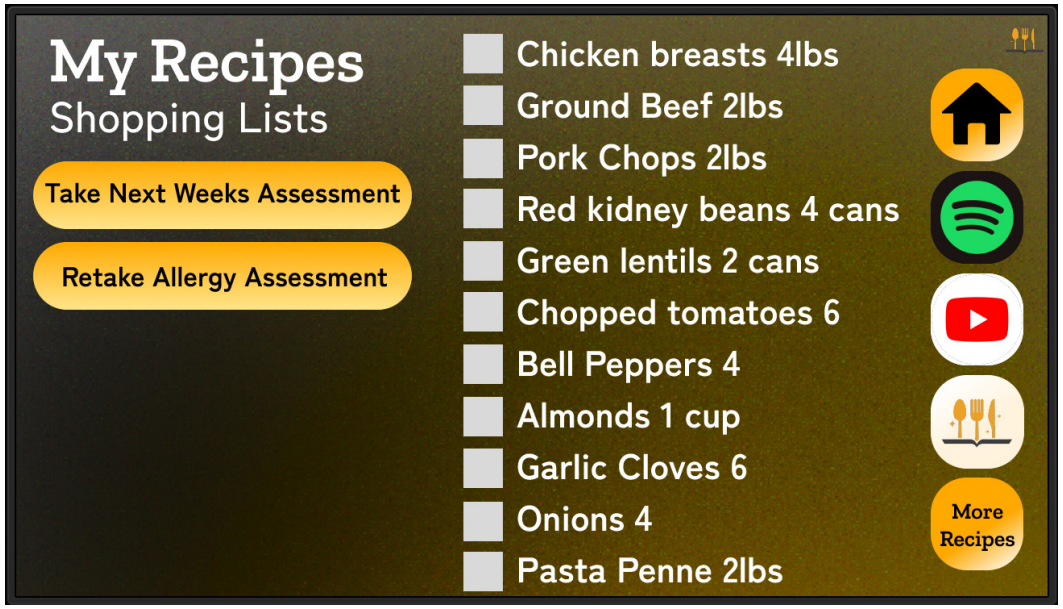


When you scroll down there is a section to the next weeks shopping list. Here you will find the shopping list for all of the recipes that you will use next week.





Here you can also take an assesment to prepare for the next week and have recipes that fit your liking and your dietary needs.



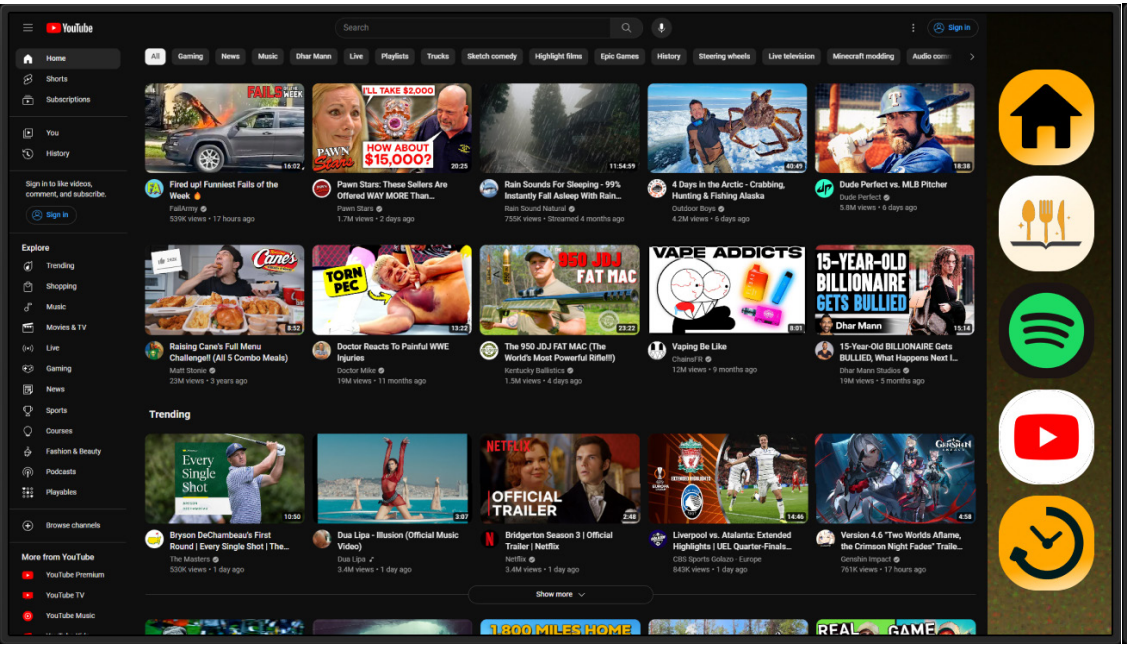
The app takes care of the shopping list for the week so you dont have to stress about not having any ingredients or not being prepared for the week.



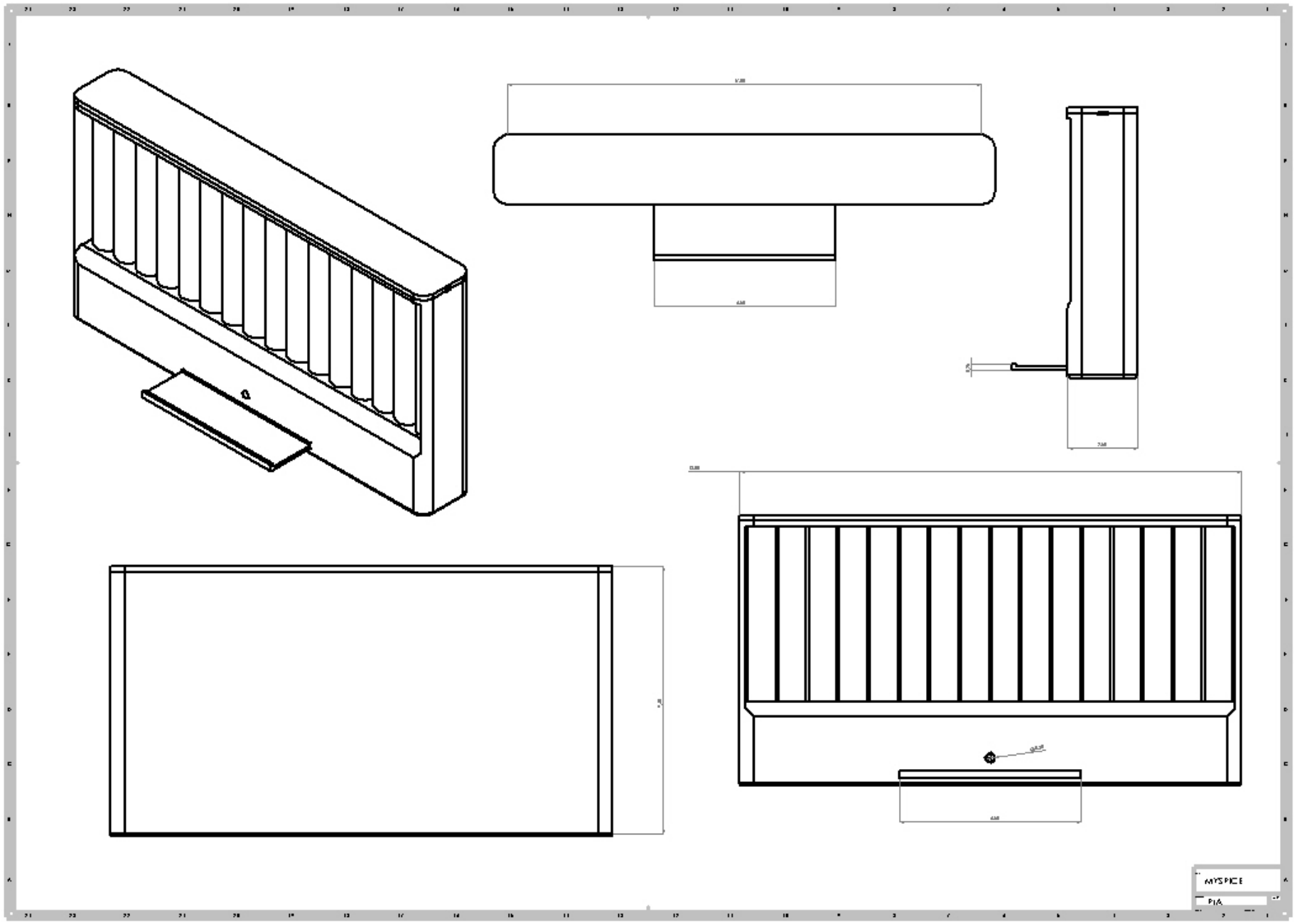


Throughout the whole process you also get access to the timer feature which allows you to set timers for your different cooking appliances without having to touch anything or spread mess anywhere.

Lastly there is of course the easy access to entertainment applications like Youtube or Spotify. While at the same time allowing you easy access to all of the features you need to navigate your recipes and meal making

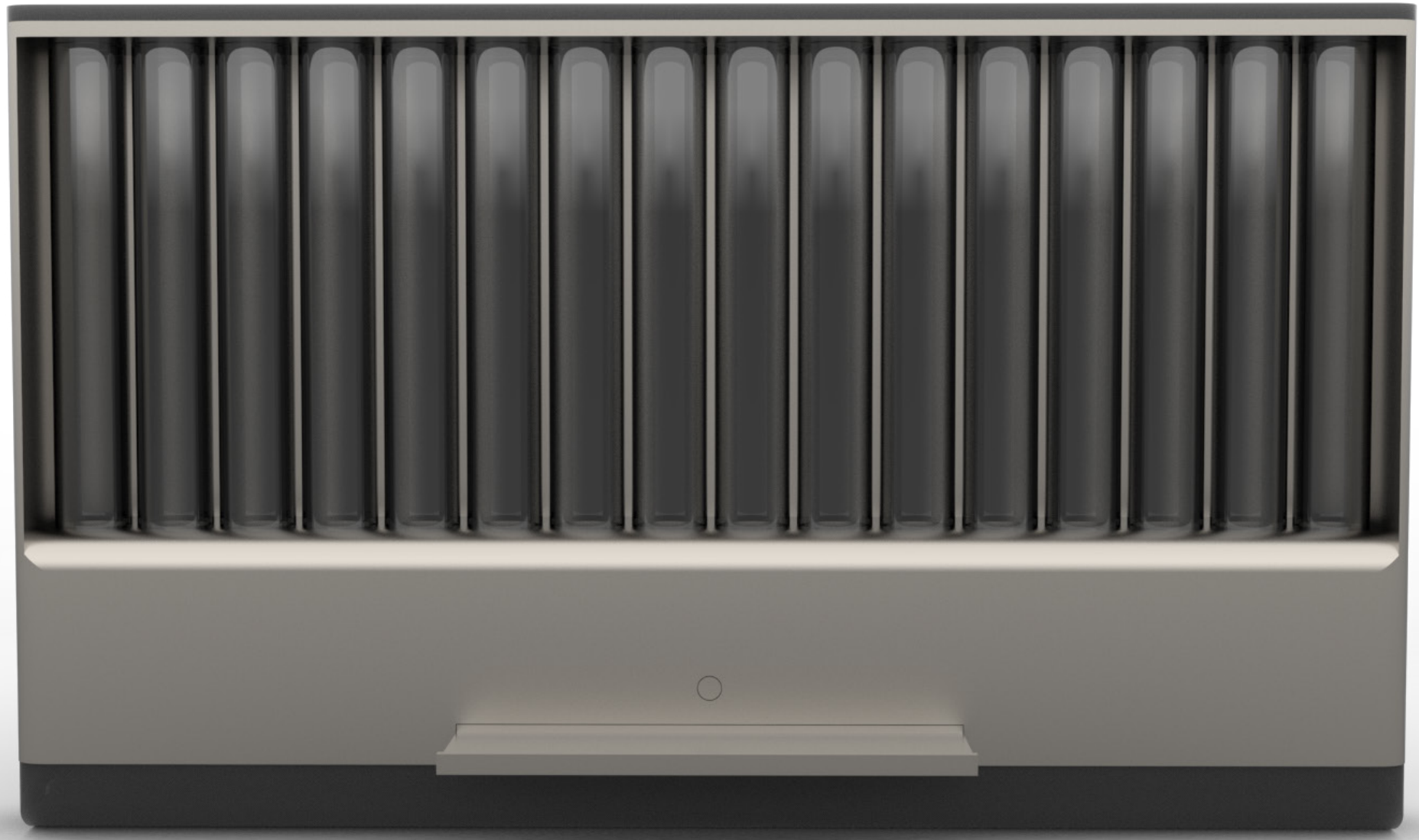




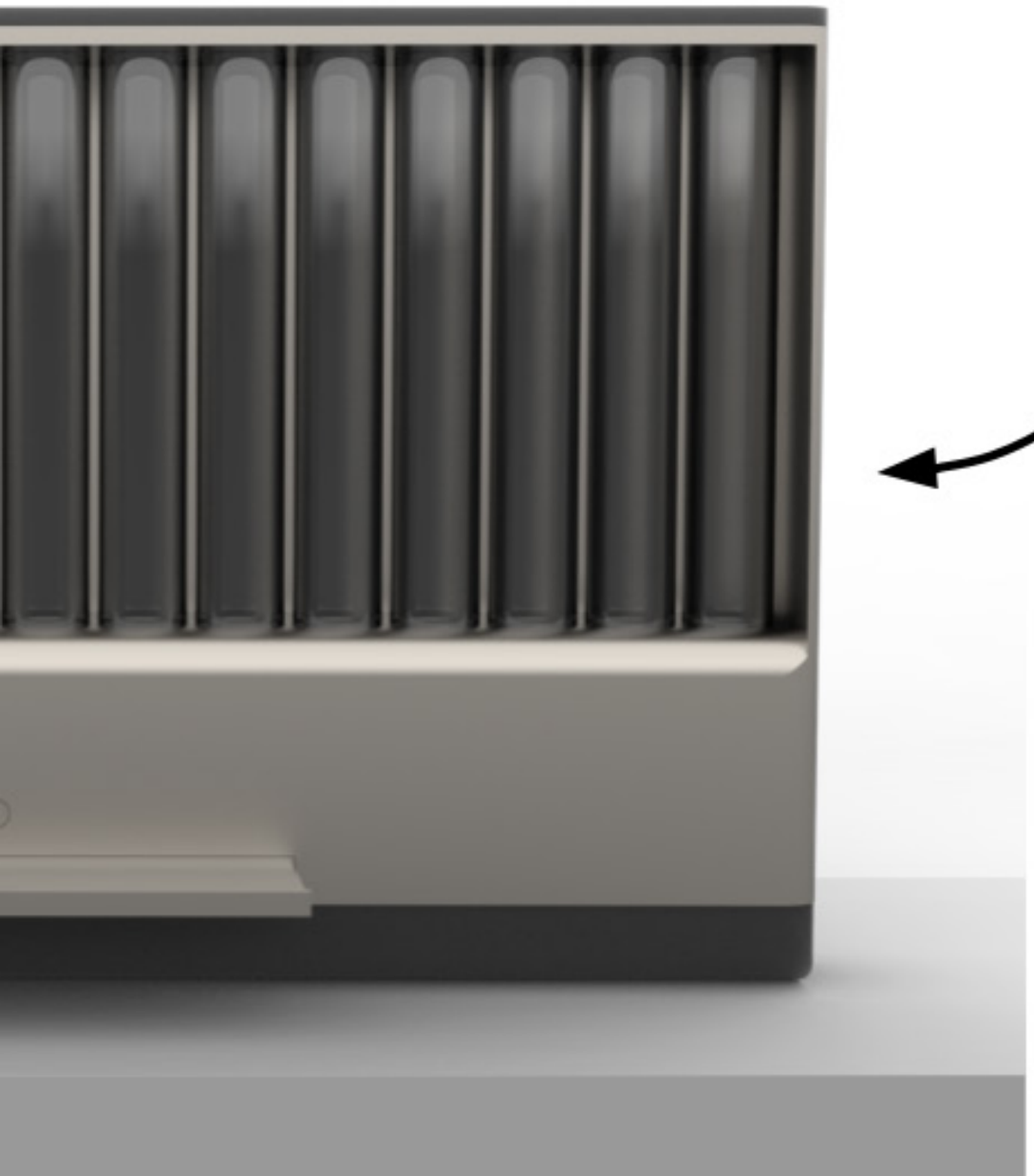


05. FINAL MODEL MAKING

I made my model using solidworks and rendered it using keyshot, I then made slides highlighting each feature for spice dispense



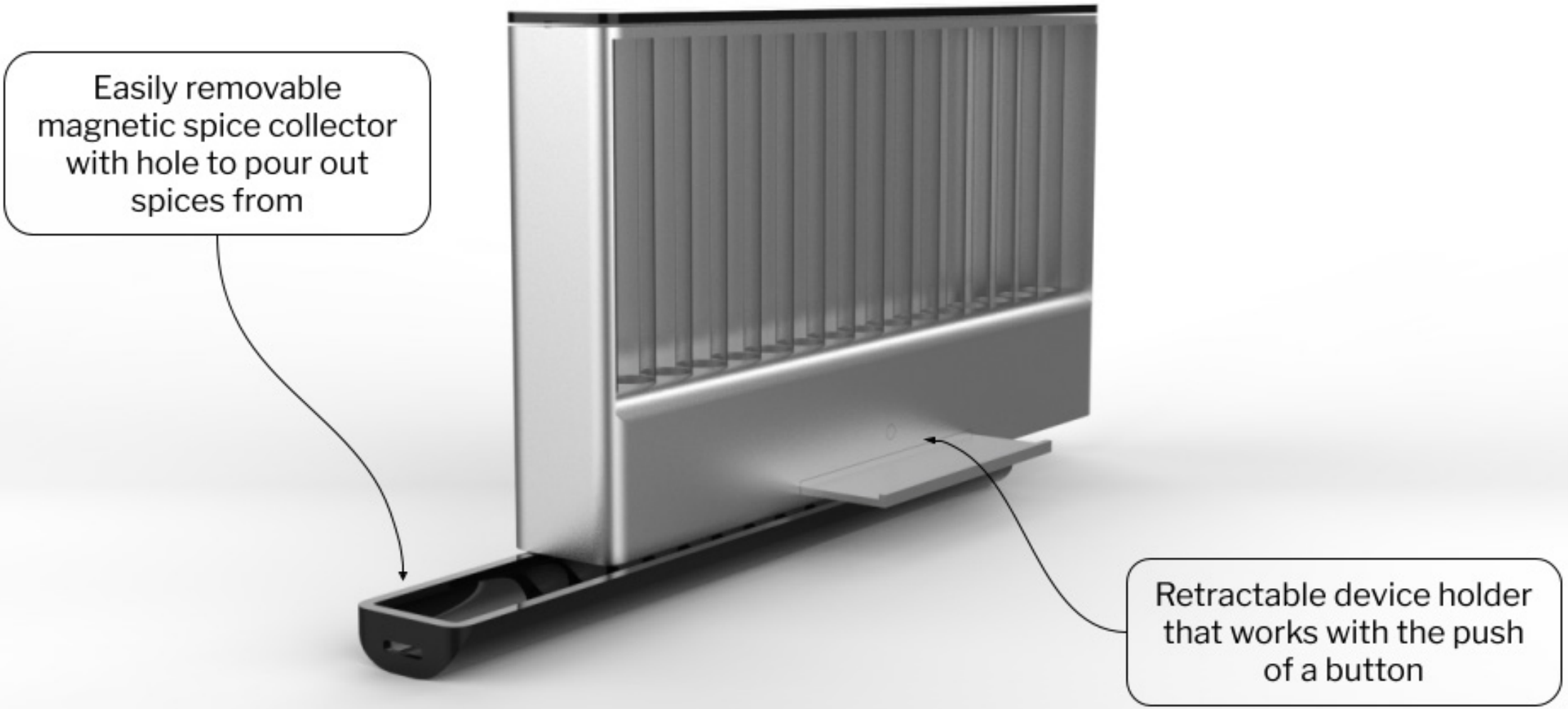
FEATURES



Clear see-through pipes to safely hold your spices while allowing you to gage how much you have left before needing to refill


Removable lid for
easy refilling of each
of your spices

FEATURES


Easily removable magnetic spice collector with hole to pour out spices from
Retractable device holder that works with the push of a button

IN-CONTEXT RENDERS



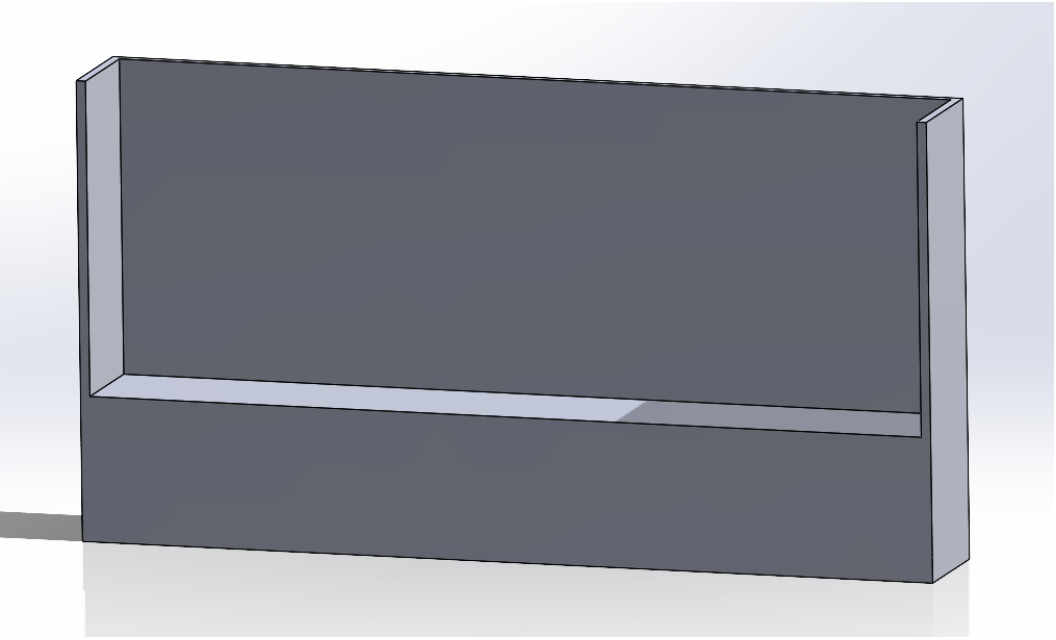


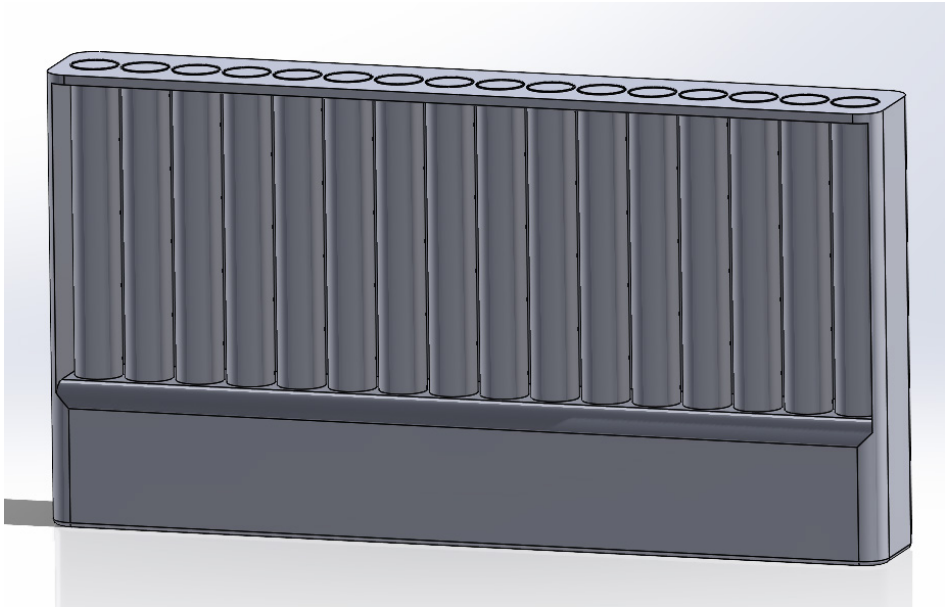


06. PHYSICAL MODEL

SOLIDWORKS

My process started by modeling in solidworks, I took the basis from my first prototype and model. That version that was featured earlier on the book was a little to open and free, I wanted the MySpice experience to be one that felt refined. So I decided to add uper walls as a casing since it made the whole model look more fine and elegant.



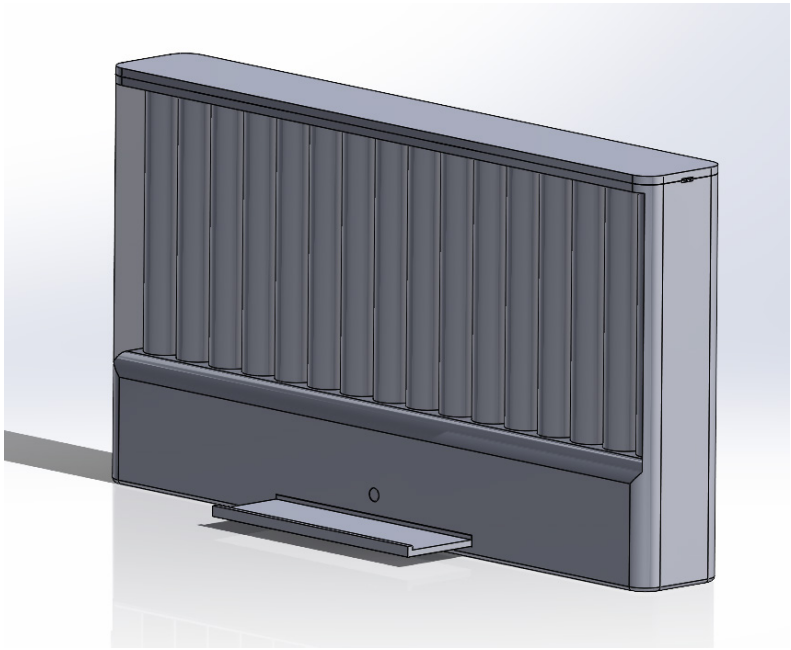


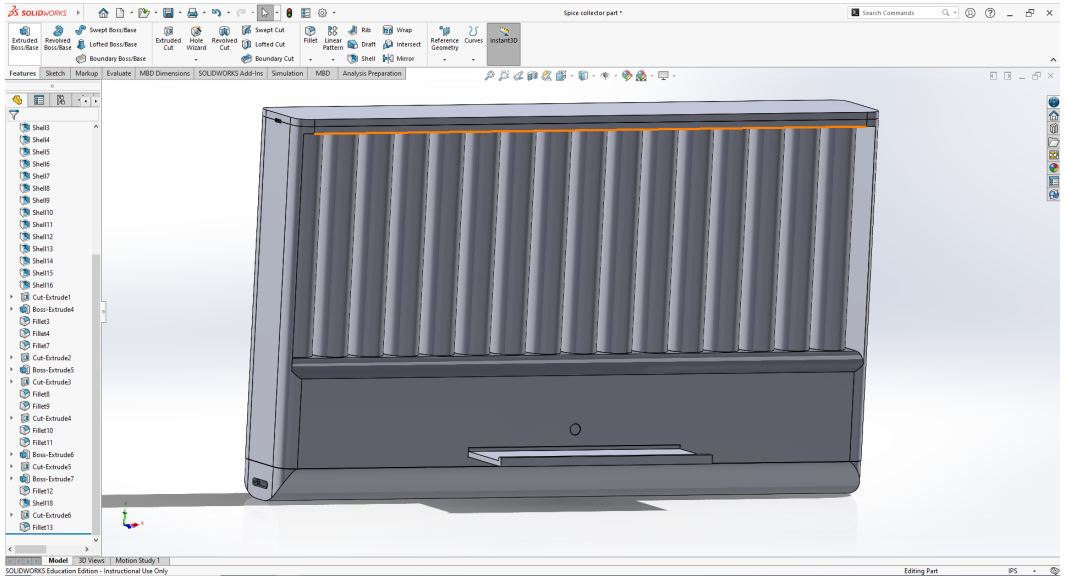
I then added curved fillets and rounded edges, because it is easier for cleaning so it made sense for a kitchen appliance, It also adds a smoother look to the product adding some visual pleasure as well as functionality.

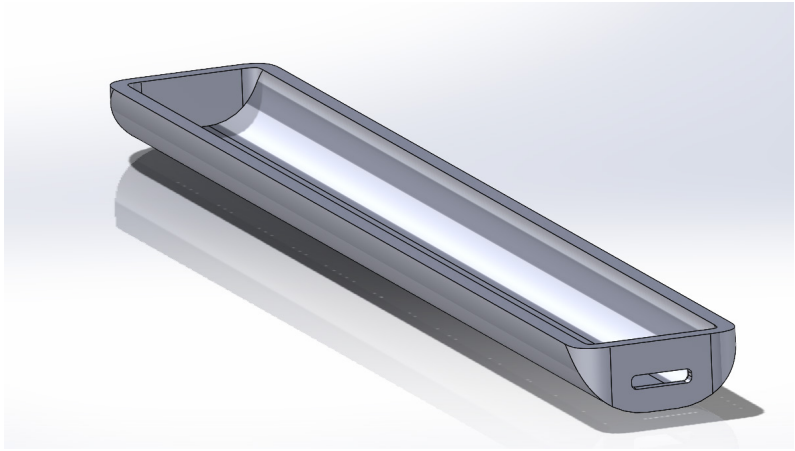
SOLIDWORKS

FEATURES

After having my chasis what was left was to add the features such as the dvice holding tray, and the spice collector that would go on the bottom. After this it was all a matter of putting it all together and then resizing it so that it would fit the printer bed and I could make my model out of PLA plastic.





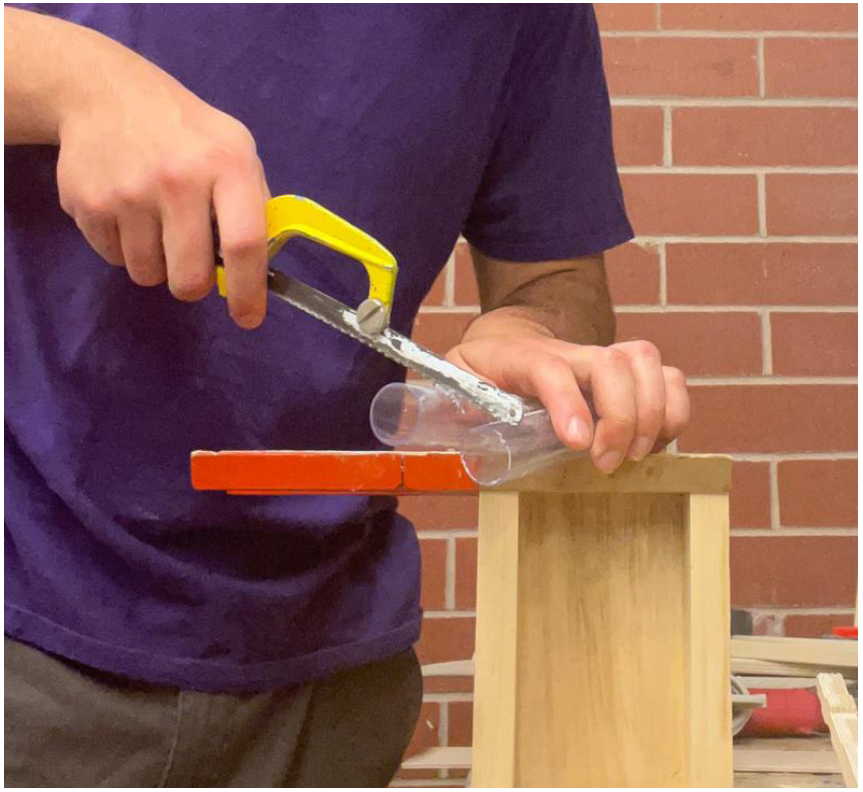


MAKING

3D PRINTING

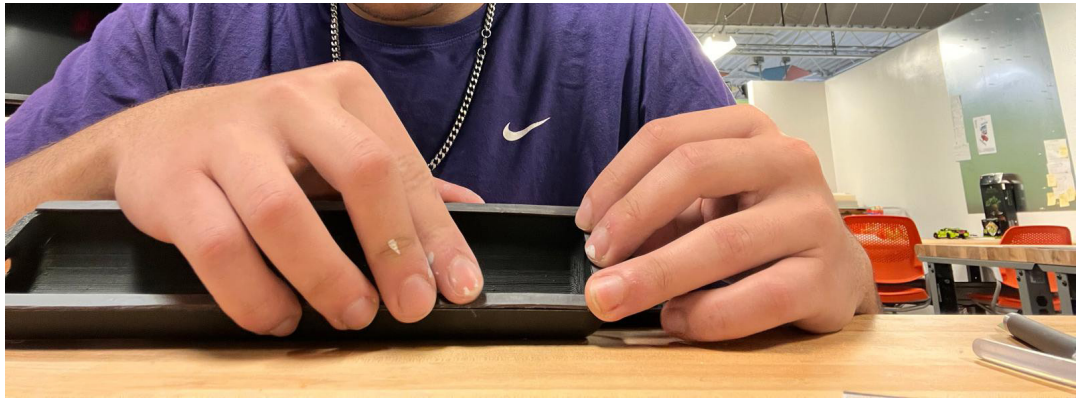
I printed all of my different parts and got them ready for making. I had my lid, spice collector and some clear PVC pipes I got from the store to keep the clear viewing. I started off by cutting all the pipes to the right measurement so they would fit my model perfectly.





MAGNETS

After having all of the elements and components ready it was time to add the magnetic aspect. I bought adhesive magnets that I added to the lid and spice collector part as well as to the main bodies so that they would be able to easily attach and de-attach from eachother.





COALESCENCE

After adding the magnets all of the components were ready to go together. I attached the lead as well as added the device holder to the chasis, then it was a matter of adding the clear pvc pipes and then the model would be totally ready!











RECIPE VIEWING
SHOPPING LIST
MEAL PLANNING
MESS-FREE
SPICE DISPENSING

MYSPICE

CREATED BY:
CARLOS OCASIO

In the end I am very content with the final product that I have created. I believe MySpice absolutely **tackles** many of the **problems in the kitchen** that were found throughout my **research**. The **time saving** aspect is a huge factor considering the **planning** that happens throughout the app, while at the same time keeping your **personal touch**. **Avoiding messiness** is another **problem that is solved** since the spice adding process is one that happens when we're usually handling foodand MySpice eliminates a lot of steps during that making it **extremely efficient**. At the same time the product is one that is **visually pleasing** and I believe can step up the environment, making it p**leasant to cook**, changing the narrative. **Instead** of cooking being **another stress factor** this device allows it to become something that you **look forward** to in your day. **Properly creating a more synchronous and holistic kitchen environment.**



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